Extended essay: commentary for example essay			
DP subject: (Subject-focused essay)	**Biology**		
DP subjects: (Interdisciplinary essay)		Interdisciplinary framework:	
Research question:	To what extent do monosodium glutamate, sucralose, carboxymethylcellulose, and sodium benzoate affect the survival of *Lactobacillus acidophilus*?		
Assessment details			
Criterion:	Mark awarded:	Commentary: (Incorporating wording from the assessment criteria)	
A: Framework for the essay [Maximum possible mark: 6]	**4**	The research question is clear although does not mention how the survival of *Lactobacillus acidophilus* is going to be measured (does not mention inhibition zone) nor the time frame of the experiment (not clear whether it is 24 or 48 hs). The research methods are mostly suitable for the research question and are explained and applied partially effectively. The experiment is on the effect of additives on *Lactobacillus*, but probiotic pills of unknown composition are used (it is not clear that they are 100% *Lactobacillus*). The Petri dishes are divided into 3 sections, it is not clear if the same additive is used per dish or if each disc contains a different additive. The text does not describe how the inhibition zone is measured, nor what happens when the zone is not a perfect circle. The length of the experiment is not clear; it is says 24 hs and 48 hs. Very good essay structure; easy to read. Good flow between introduction, methodology and experimentation results, analysis and discussion. The raw data should have been shown in an annex. Data tables, diagrams and graphs are clearly understood and appropriately labelled, and data is clearly processed.	
B: Knowledge and understanding [Maximum possible mark: 6]	**5**	Uses appropriate scientific sources to present the use of additives in food and their effect on gut bacteria. Scientific terminology is used accurately and consistently throughout the essay. They mention bacteriostatic effect of additives on bacteria but do not define this term nor do they explain why this effect is considered and why bacteriolytic effect is not contemplated. The concepts on the use of additives are clear, but a more detailed explanation of the effects of these on gut bacteria is missing. The synergistic effects of additives is going to be studied, but there is not much explanation on this effect.	

C: Analysis and line of argument [Maximum possible mark: 6]	**5**	The analysis is effective and consistently produces relevant findings. The diameter of the disc was not taken into consideration when analysing the results, but this did not affect the final outcome, as the discs were of the same size throughout. The results are tested with a t-test to see the significance in the differences. There is a clear and sustained line of argument, linking the research question with the introduction, results and conclusions.
D: Discussion and evaluation [Maximum possible mark: 8]	**7**	The discussion of the results is fully supported by peer reviewed scientific sources that study similar topics. The conclusion is a synthesis of the results, but should have included the statistical results too. The essay effectively evaluates not only the methodology used but also the sources consulted. The stengths and limitations are explained in detail.
E: Reflection [Maximum possible mark: 4]	**4**	Reflection is consistently evaluative, explaining the need to consult sources to find an explanation for discrepancies with the hypothesis proposed. The growth of the candidate is seen as they read more scientific papers, becoming more interested in the topic and being able to discuss the results obtained.
Total marks awarded	25/30	
Holistic comment on the essay:	The student demonstrates effective research skills, resulting in a research question that can be explored within the scope of the effect of additives in food on the gut microbiota. There is effective engagement with relevant research areas, methods and sources. There is excellent knowledge and understanding of the use of additives in the wider context of Biology and there is an effective application of source material and correct use of subject-specific terminology and concepts that further supports this. More detail on the effect of additives at the cellular level of bacteria were expected. Conclusions are consistent, relevant and accurately analysed, but the statistical test results are not included. Arguments show sustained reason and are supported by evidence from appropriate scientific papers. Research is critically evaluated, with strengths and limitations being presented. Coherent and consistent structure support readability of the essay, with minor flaws. Reflection on the extended essay learning experience is consistently evaluative and includes specific examples of challenges presented and how these were overcome. The reflection shows evidence of the student's growth and transfer of learning.	